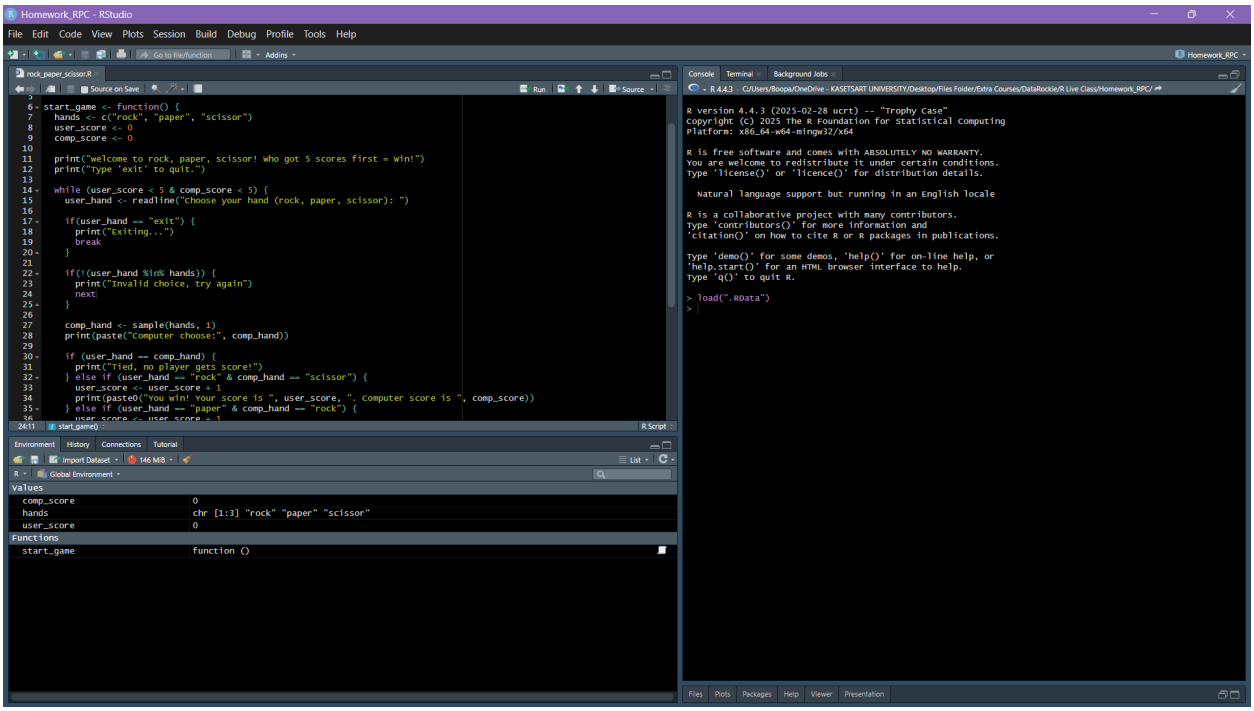
Rock, Paper, Scissor (R)

Status	Done
Assign	Pasit Promchawee
Classes	Live 06 - R Programming Foundation



Instructions

Your tasks:

- can play continuously until they are bored
- if player win get +1 point
- if draw, no one get score
- when user exit game, show summary score

Here's the sample code, this might help:

```
# random function
ABC <- c("A", "B", "C")
sample(ABC, 1)

# sample code
play_pyc <- function() {
  hands <- c("kon", "gungai", "kradad")
  while(TRUE) {
    user_hand <- readline("Choose your hand: ")
    comp_hand <- sample(hands, 1)
    if(user_hand == comp_hand)
  }
}
```

My Code

This is my whole code of rock, paper, scissor game.

```
# Rock, Paper, Scissor

library(tidyverse)


start_game <- function() {
  hands <- c("rock", "paper", "scissor")
  user_score <- 0
  comp_score <- 0

  print("welcome to rock, paper, scissor! Who got 5 scores fi
rst = Win!")
  print("Type 'exit' to quit.")

  while (user_score < 5 & comp_score < 5) {
```

```
    user_hand <- readline("Choose your hand (rock, paper, sci
ssor): ")

    if(user_hand == "exit") {
      print("Exiting...")
      break
    }

    if(!(user_hand %in% hands)) {
      print("Invalid choice, try again")
      next
    }

    comp_hand <- sample(hands, 1)
    print(paste("Computer choose:", comp_hand))

    if (user_hand == comp_hand) {
      print("Tied, no player gets score!")
    } else if (user_hand == "rock" & comp_hand == "scissor")
{
      user_score <- user_score + 1
      print(paste0("You win! Your score is ", user_score, ".
Computer score is ", comp_score))
    } else if (user_hand == "paper" & comp_hand == "rock") {
      user_score <- user_score + 1
      print(paste0("You win! Your score is ", user_score, ".
Computer score is ", comp_score))
    } else if (user_hand == "scissor" & comp_hand == "paper")
{
      user_score <- user_score + 1
      print(paste0("You win! Your score is ", user_score, ".
Computer score is ", comp_score))
    } else {
      comp_score <- comp_score + 1
      print(paste0("Computer wins! Your score is ", user_scor
e, ". Computer score is ", comp_score))
```

```
    }

    if (user_score == 5) {
      print("Congratulations! You win the round!")
    } else if (comp_score == 5) {
      print("Game Over! Computer wins the round!")
      print("Don't give up, you can try again!")
    }
  }
}
```

To activate, runs `start_game()` function and enjoy!

Details

Let's break down into details.

First is creating function called `start_game()`. This function requires zero argument.

```
start_game <- function() {}
```

`hand()` is variable where it stores `rock`, `paper` and `scissor` in vector. This is for computer randomly choose data in vector.

`user_score` is variable where it stores 0. The purpose is for adding a score for user.

`comp_score` is variable where it stores 0. The purpose is the same except this is for computer.

```
hands <- c("rock", "paper", "scissor")
user_score <- 0
comp_score <- 0
```

When starting the game, it shows welcome message.

```
print("welcome to rock, paper, scissor! Who got 5 scores firs
t = Win!")
print("Type 'exit' to quit.")
```

This function/game use `while` loop where it will run if `user_score < 5 AND comp_score < 5`.

And I create `user_hand`, a variable where it receives data by user input.

If user writes something that doesn't match any of `hand` variable, it shows that it's an invalid choice and the function will run `user_hand` again to receive the valid input.

```
while (user_score < 5 & comp_score < 5) {
  user_hand <- readline("Choose your hand (rock, paper, sciss
or): ")

  if(user_hand == "exit") {
    print("Exiting...")
    break
  }

  if(!(user_hand %in% hands)) {
    print("Invalid choice, try again")
    next
  }
```

And this game allows you to quit anytime by typing `exit`. It will break the loop

We need computer to random the hand. So, I create `comp_hand` variable where it uses `sample()` to randomize.

```
comp_hand <- sample(hands, 1)
print(paste("Computer choose:", comp_hand))
```

This also show what computer chooses.

Now, we have `user_hand` and `comp_hand`. Let's see who will get the score.

From now on, it's literally `if-else` function where I need to see who will win.

Rock, Paper and Scissor scenarios:

- Rock → Scissor = **Win**
- Paper → Rock = **Win**
- Scissor → Paper = **Win**
- If both hands are the same = **Draw**
- Else = **Lose**

We will apply to the code exactly the same as scenario.

```
if (user_hand == comp_hand) {
  print("Tied, no player gets score!")
} else if (user_hand == "rock" & comp_hand == "scissor") {
  user_score <- user_score + 1
  print(paste0("You win! Your score is ", user_score, ". Comp
uter score is ", comp_score))
} else if (user_hand == "paper" & comp_hand == "rock") {
  user_score <- user_score + 1
  print(paste0("You win! Your score is ", user_score, ". Comp
uter score is ", comp_score))
} else if (user_hand == "scissor" & comp_hand == "paper") {
  user_score <- user_score + 1
  print(paste0("You win! Your score is ", user_score, ". Comp
```

```
uter score is ", comp_score))
} else {
  comp_score <- comp_score + 1
  print(paste0("Computer wins! Your score is ", user_score,
". Computer score is ", comp_score))
}
```

If the logic is true (which means user wins) then user gets a score.

Score will add +1 to `user_score`.

If the logic is false (which means user didn't match any logics) then computer gets a score.

Score will add +1 to `comp_score`.

If both hands are the same, no score for both player.

This game, who has reaches 5 score first will win the round.

So, I add this logic.

```
if (user_score == 5) {
  print("Congratulations! You win the round!")
} else if (comp_score == 5) {
  print("Game Over! Computer wins the round!")
  print("Don't give up, you can try again!")
}
```

If `user_score == 5`, user wins.

But if `comp_score == 5`, user loses.

And here's the code detail. Let's take a look at gameplay.

Gameplay

Start a game

```
> load(".RData")
> start_game()
[1] "welcome to rock, paper, scissor! Who got 5 scores first = Win!"
[1] "Type 'exit' to quit."
Choose your hand (rock, paper, scissor):
```

User gets a score

```
Choose your hand (rock, paper, scissor): scissor
[1] "Computer choose: paper"
[1] "You win! Your score is 1. Computer score is 2"
Choose your hand (rock, paper, scissor):
```

Computer gets a score

```
Choose your hand (rock, paper, scissor): rock
[1] "Computer choose: paper"
[1] "Computer wins! Your score is 0. Computer score is 1"
Choose your hand (rock, paper, scissor):
```

Tied

```
Choose your hand (rock, paper, scissor): paper
[1] "Computer choose: paper"
[1] "Tied, no player gets score!"
Choose your hand (rock, paper, scissor):
```

User wins

```
Choose your hand (rock, paper, scissor): rock
[1] "Computer choose: scissor"
[1] "You win! Your score is 5. Computer score is 3"
[1] "Congratulations! You win the round!"
```

Computer wins

```
Choose your hand (rock, paper, scissor): rock
[1] "Computer choose: paper"
[1] "Computer wins! Your score is 2. Computer score is 5"
[1] "Game Over! Computer wins the round!"
[1] "Don't give up, you can try again!"
```

Exit a game

```
> start_game()
[1] "welcome to rock, paper, scissor! Who got 5 scores first = Win!"
[1] "Type 'exit' to quit."
Choose your hand (rock, paper, scissor): exit
[1] "Exiting..."
```